

***Autonomous UAV System For Real-Time Human
Detection and Health Vitals Monitoring***

Project Work II

Department of ECE



Sl. No.	Reg. No.	Student Name
1	21ETEC004016	BRIAN CHARLES
2	22ETEC004011	BHANU SIMHA REDDY
3	22ETEC004048	KIRAN SHINDE
4	22ETEC004051	SIMHADRI THANMAI

**Supervisors : Mr. Viswanath K,
Assistant Professor, Department of ECE**

June– 2026

**FACULTY OF ENGINEERING AND TECHNOLOGY
M. S. RAMAIAH UNIVERSITY OF APPLIED SCIENCES
Bengaluru -560 054**

FACULTY OF ENGINEERING AND TECHNOLOGY



Certificate

*This is to certify that the Project Work titled “**Autonomous UAV System For Real-Time Human Detection and Health Vitals Monitoring**” is a Bonafide record of the work carried out by **Mr. BRIAN CHARLES**, Reg. No. **21ETEC004016**, **Mr. BHANU SIMHA REDDY**, Reg. No. **22ETEC004011**, **Mr. KIRAN SHINDE**, Reg. No. **22ETEC004048**, **Ms. SIMHADRI THANMAI**, Reg. No. **22ETEC004051**, in partial fulfilment of requirements for the award of **B.Tech** Degree of M. S. Ramaiah University of Applied Sciences in the Department of **Electronics and Communication Engineering***

June – 2026

Supervisor: Mr. Viswanath K
Assistant Professor, Department of ECE

Dr. T. Christy Bobby
Professor and Head,
Department of ECE

Dr. Sarat Kumar Maharana
Dean,
FET, MSRUAS

Examiners:

Name

Signature

1

2.

Declaration

Autonomous UAV System For Real-Time Human Detection and Health Vitals Monitoring

The project work is submitted in partial fulfilment of academic requirements for the B. Tech. Degree of M. S. Ramaiah University of Applied Sciences in the Department of ECE This project work is a result of our own work and is in conformance to the guidelines on plagiarism as laid out in the University Student Handbook. All sections of the text and results, which have been obtained from other sources are fully referenced. We understand that cheating and plagiarism constitute a breach of university regulations, hence this project report has been passed through plagiarism check and the report has been submitted to the supervisor

Sl. No.	Reg. No.	Student Name	Signature
1	21ETEC004016	BRIAN CHARLES	
2	22ETEC004011	BHANU SIMHA REDDY	
3	22ETEC004048	KIRAN SHINDE	
4	22ETEC004051	SIMHADRI THANMAI	

Date: 3 June 2026

Acknowledgements

We take this opportunity to express our sincere gratitude and appreciation to **Ramaiah University of Applied Sciences**, Bengaluru for providing us an opportunity to carry out our group project work.

It brings a sense of privilege to associate this work with **Mr. Viswanath K**, our project supervisor. Working under his guidance has been a precious opportunity. We are immensely grateful for his vigilant supervision, constant encouragement, inspiring ideas, constructive criticism, and timely expertise, which reflects in the final outcome of this work.

We are grateful to the University and thank the Head of Department of Electronics and Communication Engineering, **Dr. T. Christy Bobby** for providing us with this opportunity, guidance, and encouragement throughout.

We would like to earnestly thank **Dr. Sarat Kumar Maharana**, the Dean, Faculty of Engineering and Technology, for facilitating academic excellence in the college that helped in completing this project.

We are also thankful to **Prof. Kuldeep Kumar Raina**, the Vice Chancellor, for facilitating academic excellence in the college that helped in completing this project.

Furthermore, we take this opportunity to acknowledge and appreciate the support we got from our mentors, faculty members, Lab Assistants for providing all the facilities and resources for the successful completion of the project

Finally, we are thankful for the constant support shown by our family, friends and well-wishers, their unwavering support has been a source of motivation throughout this journey

Abstract

Through this project, a Vitals Monitoring System for the drone equipped with an Autonomous Contactless Monitoring Platform to monitor the person in the disaster-affected zone to deliver real-time information to the Search and rescue teams. The system is a working prototype that will help rescue teams in finding victims; in assessing the victim's condition at a distance; and in reducing danger when approaching the individual for face-to-face assistance in dangerous situations. The system would therefore provide an efficient and effective response in the case of emergencies such as autonomous navigation, victim detection and real time health monitoring.

The prototype is built using a raspberry pi 5, and a Pixhawk flight controller. This system achieves real time victim assessment by integrating YOLOv8 Pose model for human distress recognition with MediaPipe face mesh for Remote photoplethysmography (rPPG) By applying Plane Orthogonal To Skin (POS) algorithm. The contact between the raspberry pi and Pixhawk is handled by a MAVlink protocol allowing a telemetry to transfer valuable data to external application such as telegram and an android app using WebSockets

The results of the project indicates that the proposed system can detect human body, approach to the victim, gather health information without touching the victim. The system maintained an impressive **91.35% accuracy** at a 1-meter range, which further scaled up to **91.76% accuracy** at 2 meters as the frame stabilized. **Environmental Adaptability:** Under optimal, direct light conditions, the model achieved a peak accuracy of **93.61%** with a minimal percentage error of **6.39%**. **Stress Resilience:** Even under challenging adverse conditions such as harsh backlighting (84.27% accuracy) and physical target sweat/moisture conditions (84.60% accuracy) the rPPG framework reliably isolated physiological markers well above standard emergency thresholds.

Table of Contents

Certificate	(ii)
Declaration.....	(iii)
Acknowledgements.....	(iv)
Abstract	(v)
Table of Contents.....	(vi)
List of Tables.....	(x)
List of Figures.....	(xi)
Nomenclature.....	(xii)
Abbreviations and Acronyms.....	(xiii)
Chapter-1: Introduction.....	01
1.1 Significance of this Topic.....	02
1.2 Motivation and Need for the Project.....	03
Chapter-2: Literature Review and Problem Formulation.....	04
2.1 Techniques for Human Detection.....	04
2.1.1 Gesture / Pose Detection Methods.....	04
2.2 Challenges in Real-Time Detection Systems.....	05
2.3 Introduction To Vitals Detection.....	05
2.3.1 Remote Photoplethysmography	06
2.3.2 Existing Research and Development Studies.....	07
2.3.3 The POS (Plane-Orthogonal-to-Skin) Algorithm.....	08
2.3.4 The Formula and Step-by-Step Implementation.....	08
2.3.5 Limitations of particular studies	10
2.3.6 Research Findings.....	10
2.3.7 Research Gaps.....	11
2.4. Introduction to Android Mobile Application for Autonomous Contactless Vital Signs Monitoring.....	12
2.4.1 Flutter Framework.....	12
2.4.2 Dart Asynchronous Model and State Management.....	12
2.4.3. Real-Time Communication via WebSocket Protocol.....	13
2.4.4. Real-Time Physiological Data Visualisation.....	13
2.4.4.2 fl_chart and Animated Vital Cards.....	13
2.4.5 Drone Tracking And map display	14

2.4.6. Health Status Classification and Alert System.....	14
2.4.6.1 Clinical Basis for Thresholds.....	14
2.4.6.2 Alert Mechanism.....	14

Chapter-3: Problem Statement

3.1 Title	15
3.2 Aim.....	15
3.3 Objectives.....	15
3.4 Methods and Methodology/Approach to attain each objective	16

Chapter-4: Problem Solving (Experimental, Mathematical, Statistical, Clinical approaches)

18

4.1 Hardware Architecture and Component Selection.....	18
4.2 Engineering Calculations and Propulsion Analysis.....	19
4.2.1Weight Budget Estimation (All-Up Weight).....	19
4.2.2 Thrust-to-Weight Ratio Calculation.....	19
4.2.3 Battery C-Rating and Current Safety Check.....	20
4.2.4 Battery Safety.....	20
4.2.5 Theoretical Flight Time Calculation.....	20
4.3 Hardware Integration and Wiring Strategy.....	20
4.3.1 Motor Mapping.....	21
4.4 Software Pipeline of the system.....	21
4.5 Communication System.....	23

Chapter-5: Results and Discussions

25

5.1 Various Parameter Validation(hear rate)	25
5.2 Communication Performance Evaluation.....	32
5.3 Android Application Performance.....	33
5.4 Verification and Validation of Objectives.....	33
5.5 Discussion of Results.....	34
5.6 Recommendations.....	34
5.7.1 Prototype results.....	35
5.7.1 MAVlink Connection	35
5.7.2 victim detection	36

5.7.3 Ground Station	36
Chapter-6: Project Costing	38
Chapter-7: Conclusions and Future Directions	
7.2 Comparative Analysis with Existing Literature.....	39
7.2.1 Benchmark Selection.....	39
7.2.2 Key Performance Comparison Metrics	39
7.3 System Accuracy Charts.....	40
7.3.1 Performance Evaluation Under Varied Test Conditions.....	40
7.4 Suggestions for Future Directions.....	41
References	42

List of Tables

SUSTAINABLE DEVELOPMENT GOALS (SDG) addressed in the project.....	xi
Table 2.3.5 Limitations of particular studies.....	10
Table 2.4.6: Health Classification Thresholds in vitals_model.dart.....	14
Table 3.4 Methods and Methodology.....	16
Table 4.1 Hardware Architecture and Component Selection.....	18
Table 4.2.1 Weight Budget Estimation (All-Up Weigh).....	19
Table 5.1 Heart Rate and respiratory rate against light vs smart watch data of 3 minutes	26
Table 5.1.1 Heart Rate and respiratory rate with light vs smart watch data of 3 minutes	27
Table 5.1.2 Heart Rate and respiratory rate with light and sweat vs smart watch data of 3 minutes.....	28
Table 5.1.3 Comparison of Reference and Estimated Vital Signs.....	29
Table 5.1.4 Distance Accuracy of the system.....	30
Table 5.1.5 Comparison of various test variables and their accuracy.....	31
Table 5.2 Communication Performance Metrics.....	32
Table 5.3 Mobile Application Resource Utilization.....	33
Table 5.6.3 Ground Station outputs.....	36
Table 6.1 Project Costing.....	37
Table 7.2 comparison of the prototype vs existing paperwork.....	40

List of Figures

Figure 1: Contactless Detection using drone.....	1
Figure 1.1 visualization of the ground station.....	2
Figure 2.3: Smart Watch PPG.....	6
Figure 2.3.1 Remote PPG.....	7
Figure 4.4 Software stack (POS algorithm).....	8
Figure 5.2 Communication performance Analysis.....	32
Figure 5.6 Hex quarter Build.....	35
Figure 5.6.1 MAVlink Connection.....	35
Figure 5.6.2 Victim Detection.....	36
Figure 7.3.1 Performance Evaluation Under Varied Test Conditions.....	41

Nomenclature

A	Acceleration (m/s ²)
F	Force (N)
T	Temperature (K) ,(C)
N	Speed (RPM)
DoF	Degrees of Freedom
W	Track Width (m)

Abbreviation and Acronyms

Abbreviation	Full-Form
AI	Artificial Intelligence
BLDC	Brushless Direct Current
CV	Computer Vision
COCO	Common Objects in Context
BOT API	Robot Application Based Interface
GPS	Global Positioning System
HTTP	Hypertext Transfer Protocol
MAVlink	Micro Air Vehicle Link
UART	Universal Asynchronous RX/TX
UDP	User Datagram Protocol
UAV	Unmanned Aerial vehicle
Wi-Fi	Wireless Fidelity
NDMA	National Disaster Management Authority
RPI	Raspberry Pi 5
SAR	Search and Rescue
rPPG	Remote Photoplethysmography

SUSTAINABLE DEVELOPMENT GOALS (SDG) addressed in the project:

Levels: Poor:1, Good :2, Excellent:3		
SDG	Level	Justification
Good Health and Well-being		
Quality education		
Decent work and Economic Growth		
Industry, Innovation and Infrastructure		
Climate action		

Levels: Poor-1, Good-2, Excellent-3			
	PO PSO	Contribution from the project	Level
1	Engineering Knowledge: Knowledge of mathematics, engineering fundamentals, engineering specialization to form of complex engineering problems	Applying FuzzyC and Means algorithms to perform image processing tasks	K- 3
2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.	Model image segmentation problem and automatically segment the brain tumor from the MRI images.	3

3	Design/development of solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required	To learn MATLAB graphical user interface (GUI) and develop a code according to problem statement. To segment the brain tumor using combinations of clustering algorithms with some morphological operations on the image.	3
4	Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.	Project work is carried out based on the thesis. Understanding the concepts of image processing and tumors and pixels intensity. Comparison of two clustering algorithms K-means and Fuzzy to find the area of tumor and to find the patient's condition.	3
5	Modern tool usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.	Project was carried out using MATLAB and GUI using graphical user interface development environment.	3
6	The Engineer and the world: Analyze and evaluate societal and environmental aspects while solving	It is cost effective and automated process. To complete the area of tumor from the MRI images and check the patient's	3

	complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.	condition based on the area, which makes the whole process automatic.	
7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.	Project work and report followed honor code which is verified by Plagiarism check (25%) and report conforming to Industry standard	3
8	Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.	Equal and active participation is done among the team members.	3
9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences	Effective documentation is done using Latex (Overleaf) and presented using structure easy to understand Effective presentation is also prepared highlighting the novelty, design solution, result analysis and inference giving directions for further improvement	3
10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and	Scheduling and plan of action was prepared at the beginning Plan of action and implementation was recorded in diary maintained Execution was done using Cost efficient, scalable and customization. approach	2

	leader in a team, and to manage projects and in multidisciplinary environments..		
11	Life-long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change	As project is about the ongoing and upcoming technologies, further the medical image processing is applied to detect the tumors, diseases and recent covid affected lungs severity could also be found using clustering algorithms. Self learning ability improved.	3
13	PSO1: Provide students with a strong foundation in mathematics and computing along with breadth and foundational requirement in computing, science, engineering and humanities to enable them to devise and deliver efficient and safe solutions to challenging problems in Computer Science and inter-disciplinary areas.	Applied the statistics, mathematics, machine learning algorithms, and data preprocessing to predict academic performance.	3
14	PSO2: Impart analytic and cognitive skills required to develop innovative solutions for R&D, to build creative, dependable and safe products for Industry based on dynamic societal requirements motivated and nurtured by sound theoretical and practical knowledge of time tested and long	Applied Machine Learning, Data Analytics Visualization, Predictive modelling to solve agricultural and supply-chain problems.	3

	lasting principles of computer science, current tools and technologies.		
15	PSO3: Develop managerial and entrepreneurial skills inculcating strong human values along with social, interpersonal and leadership skills required for professional success in evolving global professional environments	Designed a safety application with: emergency alert, GPS tracking and real-time communication in the Women Safety Mobile Application.	3

1. Introduction

Recently, there has been a yearly occurrence of natural and man-made catastrophes, like floods, cyclones, earthquakes, fires and industrial accidents, etc. The suffrage of human lives and properties is great in these calamities. In such cases the use of strategy “Stop, Think and Observe Plan” is used by search and rescue team, however certain conditions might not be favourable to the team, for example poor visibility, unknown structures, reduced searching personnel, populated crowd affected, etc.

These obstacles in this case safety and potential delays that can pose a threat to human life need to be overcome by protecting rescue teams. As drones can go into damaging and inaccessible zones of the disaster area and 24/7 video and live pictures can be seen, it has become a useful tool in disaster management. This assists them to get the situation clearer. Most drone systems, however, need provisioning to be monitored by human beings constantly; to see if the victims from a given point are responding, if they need any speedy medications or food packages as seen in **Figure 1**.

As the incidence and intensity of disasters are increasing, the need for advanced technologies to assist in fast and effective search and rescue (SAR) is growing too. It is one of the biggest issues for rescuers to rapidly determine the health state and responsivity of victims in hazardous, inaccessible or structurally unsound locations. Traditional methods of rescue involve direct approach to people affected and thus can create a considerable risk for rescuers and cause precious medical aid to be delayed in reaching people who need it.



Figure 1: Contactless Detection using drone

1.1 Significance of this Topic

To address this situation there is a need for intelligent systems that can fly and autonomously detect a person and take important inputs related to their health from them and provide real time image and GPS coordinates with vitals detection as shown in Figure 1.1 to the SAR teams, and continuously monitor the vitals such as heart rate and respiratory rate.

New opportunities in the field of disaster response system improvement have recently emerged based on the development of Unmanned Aerial Vehicles (UAVs), computer vision, artificial intelligence and biomedical sensing. One of these advances, contactless physiological monitoring has become an attractive research field, and allows the evaluation of the physiological parameters without direct contact with the patient. This is especially important in disaster situations where access to victims could be restricted or unsafe.

The idea pointing towards an intelligent disaster-responsive platform by combining the former and latter technologies. The system is capable of victim detection, physiological monitoring by remote and real-time provision of health information to Search and Rescue workers. The drone uses Remote Photoplethysmography (rPPG) and Computer Vision algorithms to, detect and measure vital health data including Heart Rate (HR), Respiratory Rate (RR) of humans, from a safe distance.



Figure 1.1 visualization of the ground station

1.2 Motivation and Need for the Project

Extend the rescue procedures and operations, many of which are conducted by local organizations and/or the NDMA, by placing the life of the rescuers at risk when carrying out the rescue procedures.

Human operators need to be continually deployed to manually observe that drone video feed, eyes weary, and missing important data.

Image recognition techniques are used to automatically analyse images and identify humans. Drones equipped with AI can detect human beings in real-time, and amplify relevant visual data. This alleviates workload of human operators to increase the response speed. The idea was to also develop an off-line, non-interference health monitoring system that would aid them in locating potential partners that would give them current and effective medical data. In an emergency, the system may be used as a mobile support unit for health care as well as to provide real-time monitoring of sanitation—which can be displayed directly on the drone—and process onboard data in real-time without the need for it to find its way to a computer.

The implications of this research for disaster-response operations are huge, as it would enhance the efficiency, safety and effectiveness of those operations. The proposed rescue system can support the rescue team to prioritize victims based on their medical condition, use the rescue resources efficiently and expedite the EMIs without touching victims physically. Furthermore, the study has recently been added to the growing body of work in the area of smart aerial healthcare systems which autonomous drones could then be used for life-saving and humanitarian missions.

2. Literature Review and Problem Formulation

By studying earlier work, mapping of how similar problems have been handled in the past and what techniques have been used for human detection, visual analysis and research the limitations of existing systems are identified, such as manual detection of a human would take up lots of effort and time consuming, which could be very critical for such scenarios and have a high percentage of error

A few studies have focused on UAV detections including simple still situation evaluation and self-flying drone; however, the majority of these studies have encountered the problems of drone control, image enhancements/communications, limited endurance, and a few other challenges.

2.1 Techniques for Human Detection

Object detection (in flight) customized for humans in drone images is aerial human detection. The mainstream approach is CNN based detector using annotated drone images. Leading method include:

Many widely used variants of the original Redmon et al.'s YOLO (You Only Look Once) and its successors (YOLO9000, YOLOv2, YOLOv3, YOLOv4, and so on). YOLO processes the input image in a matrix with size $S \times S$, and forecasts a small fixed number of boxes in each cell. Can be executed very rapidly (in real-time), and accommodate for several individuals within the picture frame. According to Hossain & Lee, a YOLO typically has a lower recall than two-stage detectors, but has higher accuracy in classification. YOLOv2 and v3's modifications include anchor boxes and multi-scale detection that make it better suited to detecting small objects. In reality, the majority of the papers about UAV rely on YOLOv3-Tiny/YOLOv4-Tiny, which have a lighter backbone.

2.1.1 Gesture / Pose Detection Methods

After humans have been detected, their gesture or pose must be identified. Key methods include:

Skeleton-Based Recognition: Two steps - Pose Estimation: A pose estimator extracts the coordinates of the joint locations; Pose Classification: Then classify the pose.

Hand Gesture Recognition: There is some work done by researchers into hand gestures, and not the body as a whole. In this regard, hand posture and sign-language recognition by drone images require system to be adapted to both variation of viewpoint (north/south) and scale variation (sizes of the hand).

UAV vision systems: It's hard to get high accuracy even in aerial imagery where humans are only a few dozen pixels and the image is very grainy.

Change in Weather: Outdoor images may have a lot of dim lighting, shadows, glare or contrast difference (dusk or night). Since the noise of the drone's motion reduces quality, you need to be careful of the drone's speed.

Occlusion and Crowds: During disasters, people may be partially hidden behind debris or each other. Existing detectors for UAVs are primarily designed to detect single entities. Detection and pose estimation are sometimes hard with an occluded subject or in the case of crowds. Multi-person pose estimation is beneficial in ground scenarios, whereas joints estimation using aerial-views remains less well developed.

Inference on a small drone in true real-time (30 fps +) is challenging.

Data Limitation: Few big labelled datasets of aerial gestures are available. Many studies thus rely on data collected in a lab or simulated disaster site.

Even though the error message might just be a false alarm, it can be dangerous e.g. misinterpreted as someone waving instead of a fallen shadow or animal. Systems need to be precise if they are to impose noise on rescue teams to their detriment. On the flip side, missing a person (false negative) could be fatal.

2.3 Introducing to Vitals Detection

Traditional methods of blood volume pulse (BVP) measurements rely on contact based Photoplethysmography (PPG), which tracks light attenuation through vascular tissue to detect volumetric changes in blood circulation within the skin via IR or red light emitting diodes coupled with photo diodes. Every time the heart beats, blood volume increases, causing a temporary spike in light absorption. During relaxation, the volume of blood decreases, more light is detected.

Smartwatches use the same PPG technology that sends out light to the wrist tracking real-time heart rate and blood volume pulse. Smartwatches utilize photoplethysmography (PPT) (reflective) rather than transmitting light through the finger like a clinical clip. The light directed to the sensor penetrates through the skin, and some amount reflects back to a built-in photodetector as illustrated in figure 2.3.

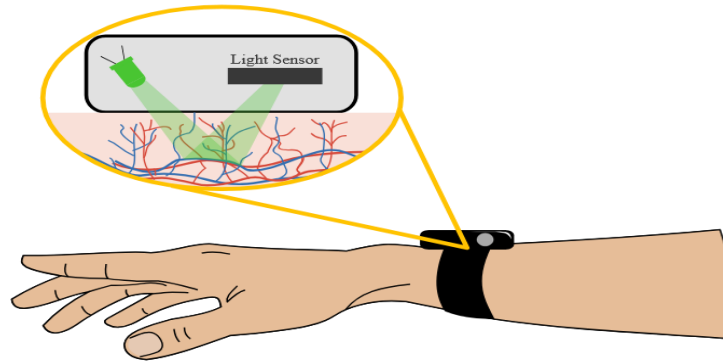


Figure 2.3: Smart watch ppg

The PPG is very accurate in a controlled clinic-like environment, but is still a contact-based technique. In any emergency and Search And Rescue (SAR) situations the operation for this contact-based method would be a sceptical approach in an autonomous drone, to favour the SAR missions, Remote Photoplethysmography (rPPG) is introduced.

2.3.1 Remote Photoplethysmography

One of the most important physiological parameters in human body, heartbeat (HR) may give a lot of information about the physical condition of a person. Unlike conventional contact method of measuring a HR, Remote Photoplethysmography (rPPG) is a method of measuring HR remotely with only camera as a contact. This non-contact measurement technique is now becoming a very interesting research field.

It has been proven that rPPG is superior providing that material subjects do not have to wear contact devices to overcome various disadvantages of contact devices, and can also be used for long term continuous monitoring. In addition, the camera used for the rPPG method is low cost and easily available, which makes it very suitable to be promoted and applied widely.

In real world scenarios, the use of rPPG methods is more difficult because of the presence of a number of factors, including lighting, facial hair, and skin tone, which may impact the accuracy of extracted rPPG signal. The signal does not agree with rPPG obtained by the traditional contact method, and this presents a challenge in processing the signal due to its lower strength.

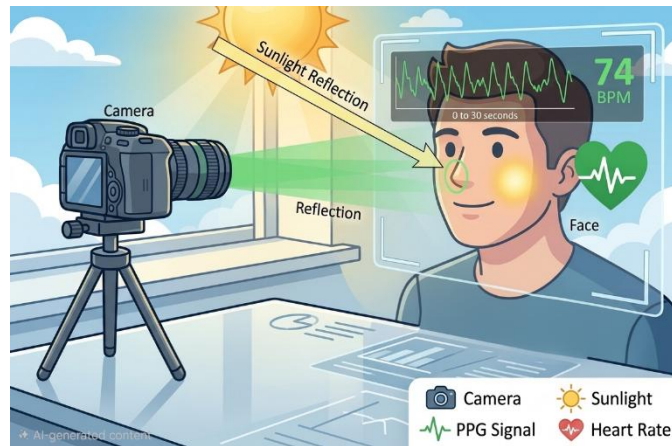


Figure 2.3.1 Remote PPG

2.3.2 Existing Research and Development Studies

With the rapid development of deep learning, new technologies in this area have spurred the emergence of many new rPPG methods for HR measurement. However, comprehensive review papers in this field are scarce, it was found that different channels of the RGB signal had varying relative strengths of PPG signals, with the green channel containing the strongest pulsatile signal. This observation is consistent with the fact that haemoglobin is most sensitive to changes in oxygenation of green light absorption, successfully demonstrating the feasibility of using rPPG methods to measure HR from ordinary consumer-grade camera footage.

Historically conventional methods often relied mainly on mathematics and algorithms, and their main purpose was to eliminate motion artifacts and noise generated in facial videos, thereby obtaining better quality rPPG signals.

The core contribution of this paper is the creation of a mathematical skin reflection model based on the **Dichromatic Reflection Model**.

- **The Unified Model:** The paper mathematically proves that the time varying camera RGB signal $C(t)$ is a mixture of three main distinct physical components:
 1. **Intensity/Luminance variations:** Light changes usually hitting all channels equally (motion induced).
 2. **Specular reflection:** Mirror like light reflection from the skin surface (contains no pulse/blood information).

3. **Diffuse reflection:** Light that penetrates the skin, interacts with haemoglobin, and contains the pulsatile PPG signal.

- **Demystifying Prior Methods:** The authors showed that popular existing methods like ICA (Independent Component Analysis), PCA, CHROM, and PBV (Blood Volume Pulse Vector) are all just different geometric strategies trying to eliminate the intensity and specular noise vectors from this unified space to leave behind the pulse signal.

2.3.3 The POS (Plane-Orthogonal-to-Skin) Algorithm

Armed with the insight that the largest source of distortion (motion-induced intensity variation) occurs along the normalized skin-tone direction, the authors designed **POS**. Instead of guessing the pulse direction using statistics (like ICA/PCA), POS completely projects the RGB signals onto a **plane that is strictly orthogonal (perpendicular) to the skin-tone vector**.

2.3.4 The Formula and Step-by-Step Implementation

The algorithm operates on a moving time window (typically 1.5 to 2 seconds long).

Step 1: Temporal Normalization

First, to eliminate the DC component (the average background skin colour and static lighting), the raw RGB signals are divided by their temporal mean over the window:

$$Cn(t) = C(t) / \mu C$$

Where **C(t)** is the raw RGB vector at a specific frame, and **μC** is the average RGB value over that entire time window.

Step 2: Projection Matrix (Orthogonal to Skin Tone)

POS defines two orthogonal projection axes on the plane perpendicular to the uniform skin-tone vector $1 = [1, 1, 1]^T$

$$X(t) = CnR(t) - CnG(t)$$

$$Y(t) = CnR(t) + CnG(t) - (2 * CnB(t))$$

Where **CnR**, **CnG**, **CnB** are the normalized Red, Green, and Blue channels. X(t) and Y(t) represent a space where uniform intensity variations along the [1, 1, 1] axis have been completely cancelled out.

Step 3: Alpha Tuning (Stepwise Tuning)

To suppress the remaining specular (mirror like) motion distortions, the algorithm mixes X(t) and Y(t) using a dynamic scaling factor alpha. The ratio of their standard deviation's sigma defines alpha:

$$\text{alpha} = \text{std}(X) / \text{std}(Y)$$

The raw pulse segment h(t) for the current window is computed as:

$$h(t) = X(t) - (\text{alpha} * Y(t))$$

Where **std(X)** and **std(Y)** represent the standard deviation of the X and Y signals over the window, and **h(t)** is the extracted raw pulse segment.

Step 4: Overlap-Add Construction

Because this calculation is done window by window, the final, continuous rPPG signal P(t) is constructed by stitching the overlapping window segments together using an overlap-add method:

$$P(t) = P(t) + h(t) - \mu h$$

Where **P(t)** is the continuous final rPPG signal, and **μh** is the mean of the current window segment used to smoothly stitch them together.

2.3.5 Limitations of particular studies

remote-photoplethysmography (rPPG) algorithms like CHROM and POS revolutionized camera-based pulse extraction, they are fundamentally constrained by their reliance on rigid geometric assumptions and linear algebra.

Limitation Factor	CHROM Algorithm	POS Algorithm
Illumination Constraints	Fails under non-white lighting (e.g., warm indoor lamps or coloured LEDs) due to rigid, pre-calculated colour weight assumptions.	Handles varying light colours better, but completely fails if the face is hit by multiple distinct light sources simultaneously.
Skin Tone Diversity	Highly sensitive to demographic variations; uses a fixed "average" skin-tone matrix that performs poorly on deeper complexions.	Dynamic normalization adapts to the user's specific skin tone, but still suffers from low Signal-to-Noise Ratio (SNR) on high-melanin skin.
Motion Profiles	Vulnerable to facial rotations, as angular movements alter the specular reflections across the red and blue channels unpredictably.	Susceptible to an "Alpha-Tuning" flaw ; certain complex head movements cause noise components to align in-phase, amplifying the artifact rather than cancelling it.
Processing & Latency	Operates continuously but is prone to baseline drift when motion artifacts shift abruptly over time.	Relies on short, moving time windows (1.5–2s), which can introduce stitching artifacts and phase discontinuities during sudden movements.
Degrees of Freedom	Limited to cancelling out a maximum of two independent environmental distortions at any given moment.	Shares the exact same mathematical bottleneck; cannot isolate the pulse if talking, rotating, and ambient flickering occur together.

2.3.6 Research Findings

We note that the green spectrum (approx. 555 nm) in general yields the highest modulation depth of the whole spectrum for remote cardiovascular tracking in an opto physiological approach. Spectrum infrared (IR) light can enter deeper into the human body.

The theoretical essence of the POS framework is based on the assumption that the reflection that can be pictorialized from the skin could be separated into two vectors, which are known as specular and diffuse vectors:

Diffuse Reflection S1: Reflects back from sub-dermal capillaries, obtains the actual volume change of the blood Reflection

Specular Reflection S2: Reflects off the top layer of the skin, reflects environmental motion, shadows and drone vibrations.

Based on math considerations, the projections are:

$$S1 = G - B$$

$$S2 = G + B - 2R$$

These two signals are mathematically orthogonal to each other, as high frequency noise in the specular channel S2 is orthogonal (not phase correlated with) the heart rate signal in channel S1, therefore it can be fully extracted and cancelled. The two signals are indeed mutually orthogonal to each other – the high frequency noise in the specular channel S2 is orthogonal (not phase correlated with) the heart rate signal in channel S1, making it possible to mathematically remove the high frequency noise without distortion and without polarizing.

Global dark current (GDC) or local and global dark current (LDC) errors are static errors caused by the absolute ambient brightness or by the distance, calculated as the ratio of the dark currents between the absolute brightness of the subject and the absolute darkness of the background. When absolute brightness or distance variations exist between the object being measured and the background, this error appears as static errors, called global dark current (GDC) or local and global dark current (LDC); it is reflected by dividing the raw intensity by the respective running darkness. However, it is assumed here that there is a linear relationship. When a subject is sweating heavily or in direct harsh sunlight the camera sensor reaches the maximum saturation of the quantization (255). At this boundary limit, the relationship becomes non-linear, which reduces the accuracy of the orthogonal coordinate mapping, and adds discontinuities to the final heart rate calculation.

2.3.7 Research Gaps

- Modelling: No specific literature available that explains the high-frequency vibration of BLDC motors interfering with rPPG based camera system.

- Contactless detection: Very less literature discussed on rPPG and POS algorithms, which affects the literature survey and authentication in the process of designing the prototype as one of the first generation algorithms.
- Software Edge Constraints: The need for research into rPPG processing on low power edge devices, simultaneously processing multiple kinds of face-mesh tracking and frequency estimation.
- Outdoor Lighting: Present models are not always able to adjust smoothly in response to rapid and uncontrolled ambient light changes during an outdoor flight.

2.4. This is the start of the Android Mobile Application for Autonomous Contactless Vital Signs Monitoring.

This is Introduction to Android Mobile Application for Autonomous Contactless Vital Signs Monitoring.

Unmanned aerial vehicles (UAVs), a field that will continue to gain rapid momentum in recent years, coupled with computer vision and mobile devices, has made it possible for new, innovative methods of remote health monitoring. Drone Vitals Monitor is an Android mobile app created in this project that is the operator interface of an overall autonomous remote health monitoring system. It streams physiological data in real-time from a Raspberry Pi 5 mounted on a drone via a local WIFI WebSocket connection, feeds back live data onto charts for vital signs, plots the drone trajectory on an interactive drone map and alerts for significant physiological conditions if their measured value is seen to exceed safe thresholds. This is a unique application that combines medical monitoring elements, real-time data rendering and spatial tracking in a single mobile platform for Android smartphones, which is more than just a flight management UAV software set. It is an extraordinary application for integrating medical monitoring parts into a single mobile platform for Android Smartphones, playing out live information visualisation as well as spatial tracking which is additionally more than just a flight controlling UAV application organization.

This is a literature review of the academic and technology underpinning the application, on which only the mobile software layer: Flutter cross-platform framework, WebSocket based communication, real-time physiological data visualisation, GPS mapping, health threshold classification, user interface design principles and the researched gap tackled by this work will be addressed.

2.4.1 Flutter Framework

Flutter is an open-source UI toolkit that is based on the Dart programming language. Unlike hybrid tools like React Native, where native components are connected to application logic via a JavaScript communications bridge, Flutter's Skia rendering engine can now render all UI elements directly on a dedicated GPU canvas, offering a true hybrid-free alternative. This is an architecture that eliminates frame rate degradation as seen as part of the bridge-based frames.

2.4.3. Real-Time Data Communication Using WebSocket

2.4.3.1 WebSocket Architecture

The system uses the WebSocket protocol to provide a real time communication between the Raspberry Pi 5 and the Android application through a single TCP connection. Unlike traditional polling methods, WebSocket allows data to be sent immediately when it is available, resulting in lower delay and better bandwidth usage.

In this project, the Raspberry Pi 5 acts as a Python-based WebSocket server in JSON format. The transmitted data such as heart rate, respiratory rate, signal quality, battery level, altitude, GPS coordinates, flight mode, armed status, and timestamp.

2.4.3.2 Fault Tolerance and Data Format

Wireless communication in outdoor disaster-response environments is inherently unreliable. The system implements a 5-second auto reconnect that fires whenever the WebSocket connection drops

2.4.4 Importance of Real-Time Vital Sign Charts

Displaying vital signs in graphical form makes it easier for the SAR to plan and execute. Instead of viewing only numerical values, The application has a separate scrolling chart that update in real time and store recent readings. This helps rescue personnel quickly identify any abnormal changes in a person's condition during any emergency situations.

2.4.4.2 Chart Display and Vital Status Cards

The application uses the **fl_chart** package to display real-time graphs of physiological parameters. The charts updates automatically as a new set of information is received from the drone. Along with graphical displays, the application shows vital information using dedicated cards with progress indicators and status animations. These visual elements help the SARs understand the current health status of the victim.

2.4.5 Drone Location Tracking and Map Display

The application provides a map which is used for tracking the drone's position and victims' location during operation. The map is implemented using the **flutter_map** package with OpenStreetMap data. A dark map theme is used to improve visibility during outdoor use and reduce screen power consumption.

The data received from the Pixhawk flight controller through the MAVLink protocol is transmitted along with telemetry information and displayed in the app, containing battery status, altitude, GPS location. The application continuously updates the drone's position and draws the flight path for easy tracking.

2.4.6. Health Status Classification and Alert System

2.4.6.1 Clinical Basis for Thresholds

The application analyses the incoming vitals from the victim using predefined minimum values. Based on these limits, the readings are classified into normal, warning, or critical conditions.

Vital Sign	Normal	Warning Low	Warning High	Critical
Heart Rate	60–100 BPM	< 60 BPM	> 100 BPM	< 50 or > 120 BPM
Respiratory Rate	12–20 /min	< 12 /min	> 20 /min	< 8 or > 30 /min
Signal Quality	> 70%	50%–70%	—	< 50%
Battery Level	> 30%	20%–30%	—	< 20%

Table 2.4.6: Health Classification Thresholds in vitals_model.dart

The implemented thresholds for heart rate, respiratory rate, signal quality, and battery level help the operator quickly understand the current status of the monitored person and the drone.

2.4.6.2 Alert Mechanism

Whenever any vital parameter reaches a critical level, the application generates an alert on the dashboard. The alert is designed to red show the important level such that the sars can act upon Along with the warning message, the current GPS location of the drone is displayed for a quick identification of the victim's position and health status.

3. Problem Statement

3.1 Title

Autonomous UAV System For Real-Time Human Detection and Health Vitals Monitoring

3.2 Aim

To Design and Develop a real time Autonomous Drone with AI, remote Photoplethysmography using image processing for vital detection of human being that is capable to transmit live data seamlessly, making it easy for SAR teams to successfully complete their operations without any risk to human life.

3.3 Objectives

1. To carry out literature and review for contactless Heart rate monitoring, Android applications and Autonomous drone systems.
2. To design and develop an autonomous flying hexacopter platform with Pixhawk and mission planner.
3. To Implement image processing techniques and AIML algorithms on the Pi 5 to extract Heart Rate and Respiratory Rate via camera feed.
4. To develop an android app for visual outputs and information such as GPS location, heart rate, respiratory rate and distress detection.
5. To Test and validate the accuracy and response of the system in various conditions such as lighting and distance

3.4 Methods and Methodology/Approach to attain each objective

Table 3.4 Methods and Methodology

Objective No.	Statement of the Objective	Method / Methodology	Resources Utilized
1	Develop a drone-mounted physiological monitoring platform	Facial video acquisition using onboard camera and OpenCV-based facial detection techniques	Drone Platform, ArduCam Camera, Raspberry Pi 5, Pixhawk, GPS module
2	Extract physiological parameters using rPPG	POS Algorithm, ROI Selection, Signal Normalization, Butterworth Filtering, FFT Analysis	Python, NumPy, SciPy, OpenCV
3	Integrate onboard computation and telemetry communication	MAVLink Communication, Edge Computing Architecture, Data Aggregation Framework	Raspberry Pi 5, Pixhawk 2.4.8, MAVLink Protocol
4	Develop Android monitoring application	Flutter Development Framework, WebSocket Communication, Real-Time Visualization, BOT API	Flutter SDK, Android Studio, Dart, telegram
5	Evaluate performance and system reliability	Experimental Testing, Statistical Analysis, Accuracy Comparison, Latency Measurements	Smart watch, Android Device, Testing Environment

From table 3.4

Stage 1: System Design and Hardware Integration

Firstly, choosing the right hardware, from the drone platform to the camera sensor, companion computer (Raspberry Pi) the Pixhawk flight controller and the Android monitoring device. The integration is carried out in the hardware to ensure the communication between all the subsystems.

Stage 2: Computer Vision and Signal Processing Development

The second step would be the implementation of algorithms of Face Detection, Region of Interest (ROI) extraction and Remote Photoplethysmography. Noise from the data is filtered and physiological parameters are estimated using digital signal processing (DSP).

Stage 3 Telemetry and Communication Development.

During the third phase, and to connect to the Android monitoring app, the communications between systems / facilities on board with the Android monitoring app must be established using WebSocket wireless communication. Flight telemetry and physiological data is packaged together into structured blocks of data.

Stage 4: Development of Mobile App

The fourth phase is in charge of creating and using the Android dashboard to design and develop the app. Real-time charts, telemetry displays, map interfaces, emergency alert and continuous monitoring systems are included in the application. And a Telegram BOT API that allows to send images containing information.

Stage 5: Experimental Verification and performance Evaluation

The last step is the evaluation of the performance of the developed system in the controlled experiments. Physiological parameters are measured and compared with the standard smart watch parameters and then the parameters of communication like the Latency, Packet Loss and Responsive are studied.

4. Problem Solving

The most critical and crucial key to this chapter is how to maximize the Thrust-to-Weight Ratio to be able to include the bulky computational load (Raspberry Pi 5) without compromising the flight stability or safety.

4.1 Hardware Architecture and Component Selection

In order to guarantee a well-built platform and a powerful one that could survive outdoors and be able to process its edges, the following particular components as shown in Table 4.1 have been chosen and integrated:

Table 4.1 Hardware Architecture and Component Selection.

Hardware Architecture and Component Selection Component	Component Specification	Motive
Airframe	F450 Quadcopter Frame	High structural integrity with integrated PCB points.
Propulsion	1000KV BLDC Motors (A2212)	Provides high RPM efficiency for 10-inch propellers.
Speed Control	30A SimonK ESCs	Fast refresh rates (400Hz) optimized for multi-rotor stability.
Power Plant	5200mAh 3S LiPo (35C)	High capacity for extended flight times; 35C burst rating ensures safety.
Flight Brain	Pixhawk 2.4.8	Redundant IMU and barometer for precise GPS
Flight Modes	Flysky i6s	Loiter
Companion Computer	Raspberry Pi 5 (8GB/16GB)	YOLOv8, image processing

4.2 Engineering Calculations and Propulsion Analysis

A mathematical feasibility study was carried out with ascertaining the possibility of the selection of 1000KV motors used and subsequently 30A ESCs so that the system could be lifted safely before assembly.

Budgets of weight are calculated as a portion of segregated weight per unit (cubic meter).[3]

4.2.1 Weight Budget Estimation (All-Up Weight)

Budgets of weight are determined as a percentage of segregated weight per unit (cubic meter). To begin with, the Total Take-off Weight has been established by adding the mass component of the item:

Table 4.2.1 Weight Budget Estimation (All-Up Weigh)

Component Category	Original	New Estimate	Change
Airframe & Landing Gear	300 g	450 g	+150 g
Propulsion (Motors + ESCs)	320 g	440 g	+120 g
Energy (Battery Pack)	400 g	800 g	+400 g
Primary Logic (Pixhawk/GPS)	120 g	120 g	0 g
Companion Logic (Pi 5 + Fan)	100 g	110 g	+10 g
Sundries (Wiring, PDB, Props)	60 g	80 g	+20 g
Total Take-Off Weight	1300 g	2000 g	+700 g

4.2.2 Thrust-to-Weight Ratio Calculation

Agility and stability are critical factors to a drone. A drone that is safe needs a minimum of 2:1 Thrust to weight ratio.

Motor Spec: The 1000KV motor is of a standard A2212 type with 10 inch propellers that provides about 800g of thrust at 100% throttle.^[3]

Total Available Thrust

$$T_{max} = 6 \times 800g = 4800 g$$

Ratio calculation:

$$\text{Ratio} = \frac{T_{max}}{W_{total}} = \frac{4800}{1300} \approx 3.69:1$$

Inference: This is an excellent ratio of 3.69: 1. It will result in the drone flying at about 4045% throttle resulting in plenty of remaining power which can be used during fighting strong winds or fast maneuvers during an emergency.^[3]

4.2.3 Battery C-Rating and Current Safety Check

We have to make sure that the 30A SimonK ESCs and the 35C Battery will not burn in full load. Max Current Draught: Each motor at 100 percent throttle has a draught of about 15A.

$$I_{max_load} = 6 \times 15A = 90 A$$

ESC Capacity:

$$6 \times 30A = 180A \text{ total capacity. Since } 90A < 180A$$

the SimonK ESCs are certainly clear of the danger zone.

4.2.4 Battery Safety

$$I_{battery_max} = \text{Capacity (Ah)} \times \text{C-Rating}$$

$$I_{battery_max} = 5.2 \text{ Ah} \times 35 = 182 A$$

Conclusion: 182A can be provided by the battery, however, at best, the motors will pull up 60A making the electric circuit is also safe.

4.2.5 Theoretical Flight Time Calculation

$$\text{Time}_{flight} = \frac{\text{Usable Capacity (Ah)}}{\text{Hover Current (A)}} \times 60 \text{ min}$$

$$\text{Time}_{flight} = \frac{4.16}{17} \times 60 \times 60 \approx 876 s$$

4.3 Hardware Integration and Wiring Strategy

The power circulation was done in a Power Distribution Board (PDB).

1. PDB Integration: Battery lead (XT60) is also attached to the PDB. The four SimonK ESCs are peeled off and solder on the pads of the PDB to balance the 11.1 V equally.
2. BEC (Battery Elimination Circuit): Because the voltage is 11.1V AC the Pixhawk and Raspberry Pi are unable to read it directly, a 5V/3A BEC is needed to reduce the voltage to the flight controller.

4.3.1 Motor Mapping

- Motor 1 and 2 Counter-Clockwise (CCW)
- Motor 3 and 4 Clockwise (CW)

Poor wiring in this case can cause the drone to undergo a flip as soon as it takes off.

4.4 Software Pipeline of the system

From **Figure 4.4** it describes the process of how the POS algorithm works, having different components contributing for the working model of the system

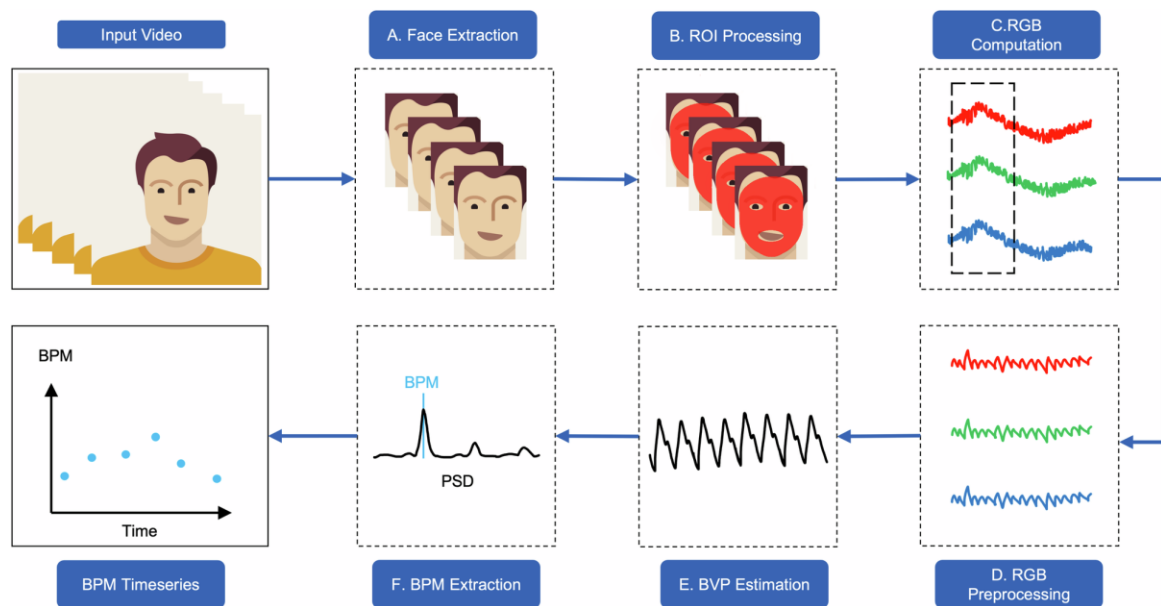


Figure 4.4 Software stack (POS algorithm)

- **Media Pipe:** 3D face mesh model - find and track the face in 3D space from raw input video.
- **OpenCV:** used to draw the HUD and process the video frames.
- **SciPy/NumPy:** used to run the heavy mathematical filters and signal analysis.

- **4th Order Butterworth Bandpass Filter:** For Heart Rate, it only allows frequencies between 0.75 Hz and 3 Hz (45–180 BPM). to block out "non-human" frequencies high-frequency drone rotor vibrations
- **Welch's Power Spectral Density:** Instead of just looking at a messy wave, Welch's method breaks the signal into segments and averages them. reduces "noise spikes" and allows the system to identify the true, dominant heart and breath rates with much higher stability than a standard Fast Fourier Transform (FFT)
- **Signal Quality Index - SQI:** A real-time self-assessment tool that compares the strength of the "Pulse" against the background "Noise." If the SQI is low, the HUD turns yellow/red to warn the operator that the data is currently unreliable due to movement or lighting.
- **Non-Linear Median Filter :** A final decision-making layer that looks at a window of the last 20 readings. If the system suddenly detects a fake jump to 200 BPM for one frame (a glitch), the Median filter discards it, keeping the display smooth and consistent

The system architecture utilizes a multi-threaded pipeline designed to maximize frame processing efficiency while keeping flight telemetry and data broadcasting running simultaneously without any lag.

I. Inputs

High-Speed Video Capture: Raw video frames with a resolution of 1280×720 are continuously captured from the Arducam of approx. 80fps

Flight Controller Telemetry: connection established between the Pixhawk flight controller using the pymavlink. enables continuous retrieval of telemetry information, including GPS latitude, longitude and altitude.

II. Processes

Human Tracking and Pose Recognition: Each incoming video frame is processed using an optimized YOLOv8-Pose model. The framework detects and tracks individuals, assigns unique tracking IDs, identifies 17 skeletal key points, and performs relative coordinate analysis to determine whether the wrists are positioned above the shoulders or not, and annotates using OpenCV for a better identification.

Forehead Region : For nearby humans, the MediaPipe Face Mesh model detects facial landmarks and isolates the forehead Region of Interest (ROI).

Remote Photoplethysmography (rPPG): The system measures variations in the average Red, Green and Blue pixel intensities over a rolling time window.

Digital Frequency Extraction: The extracted pulse signal is passed through a Butterworth bandpass filter operating between 0.75 Hz and 3.0 Hz for heart rate estimation to cut off any non human signals. The filtered signal is then analysed using Fast Fourier Transform (FFT) and Welch's Power Spectral Density techniques to estimate Heart Rate (BPM) and Respiratory Rate (RPM).

Central Thread Synchronization: A thread-safe data management mechanism combines telemetry inputs and calculated biometric parameters. The collected information is organised into a structured JSON format for real-time transmission.

III. Outputs & App Transmission

Tactical HUD Frame Rendering: OpenCV Generates a real-time annotations displaying system performance metrics, object tracking bounding boxes, and an interactive digital dial gauge for tracking active vital statistics.

Dual-Stage Telegram Alerts: An independent background thread used to instantly dispatch a localized distress alert containing live GPS coordinates and an automated 10-second compressed health triage assessment directly to remote emergency response channels.

High-Frequency WebSocket Broadcast: A network distribution engine streams the compiled JSON.

Ground station app: The WebSocket client built into the Android application captures the incoming network stream, validates the keys via a safe deserialization engine, and routes the data directly to the user interface. This instantly updates the live map tracking markers, triggers "Safe/Danger" visual alert panels, and populates the responsive heart rate dials and respiratory rate charts on the android system dashboard without any missind data.

4.5 Communication System

The system relies on a dual-protocol, multi-interface communication architecture to ensure thread-safe data handling, low-latency streaming, and long-range tactical alert routing.

I. MAVLink Telemetry Interface

Protocol Overview: Micro Air Vehicle Link (MAVLink) is an ultra-lightweight, Marshalling-based telemetry standard used to interface the high-level Companion Computer (Raspberry Pi 5) with the flight autopilot board (Pixhawk).

Implementation & Security: The connection is bound via a hardware serial UART connection. Using the pymavlink programming interface, the Pi initiates asynchronous socket polling. By decoding structured data structures (HEARTBEAT, SYS_STATUS, GLOBAL_POSITION_INT), the script extracts altitude, vehicle orientation, and GPS metrics without blocking the processing speed of the companion AI core.

II. Asynchronous WebSocket Server

Protocol Overview: WebSockets provide a full-duplex, persistent connection over a single TCP/IP connection socket, eliminating the overhead of standard HTTP request-response headers.

Implementation: The script instantiates an internal server bound to a wildcard network adapter address (0.0.0.0) running on port 8765.

Data Pipeline: Every 200 milliseconds (5 Hz streaming frequency), all available telemetry and processed data are collected using a thread-safe data lock and organized into a flat JSON payload. The generated payload is then transmitted simultaneously to all connected mobile application clients.

III. HTTP Telegram Bot API Integration

Protocol Overview: The system uses secure Hypertext Transfer Protocol (HTTPS) to communicate with system through internet.

Implementation: When a DANGER gesture is detected or a valid facial signature is successfully identified, the notification routine is triggered and sends an outbound POST request to the Telegram Bot API endpoint. This process runs on a separate execution thread, allowing the drone to transmit live images and GPS coordinates to remote disaster response teams without affecting the ongoing vision processing and telemetry operations.

5. Results and Discussions

the Autonomous UAV System For Real-Time Human Detection and Health Vitals Monitoring was tested and evaluated in a systematic manner to verify its performance and successful implementation. The primary objective of these experiments was to get the accuracy, reliability, and usability of the developed system under controlled testing conditions.

The results presented in this chapter are based on multiple experimental trials conducted using the developed prototype. The physiological parameters obtained from the system were compared with the reference medical instruments to confirm the accuracy of measurement of the system. Moreover, the latencies in communication, packet delivery performance and resource utilization of the applications were analysed to measure the appropriateness for use in real-time disaster-response applications. The results obtained are presented through tabulation and graphical representation method for better interpretation and comparisons.

5.1 Various Parameter Validation

This analysis examines the heart rate (BPM) response during three minutes by isolating specific states Active, Peak, and Recovery the data was collected under varying environmental and physical conditions that typically effect the ppg such as **Against Light, With Light, Light & Sweat**. we can quantify the efficiency of the system in transitioning from moderate activity to high-intensity stress and back to a resting state.

Table 5.1 shows the experiment conducted in 3 phases, comparing the heart rate and respiratory rate with a smart watch and testing out its accuracy

The three phases are defined as follows:

Phase 1: Active (0:00 – 1:00 min)

Initial readings are recorded during steady movement or the warm-up period. This phase establishes a baseline value to determine the starting physiological condition before intense activity begins.

Phase 2: Peak Workout (1:01 – 2:00 min)

Readings are collected during the period of maximum physical activity. This phase is used to identify the Peak Heart Rate, which indicates the body's response to high-intensity exertion.

Phase 3: Recovery (2:01 – 3:00 min)

Measurements are recorded immediately after the peak activity phase. This stage is

important for evaluating fitness levels, as it determines the Heart Rate Recovery (HRR), which represents how quickly the heart rate returns towards its baseline value.

Table 5.1 Heart Rate and respiratory rate against light vs smart watch data of 3 minutes

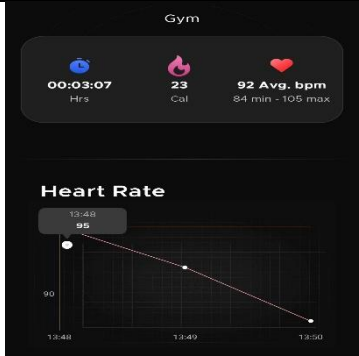
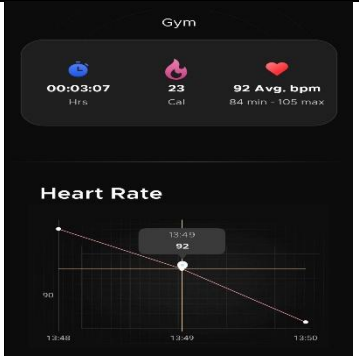
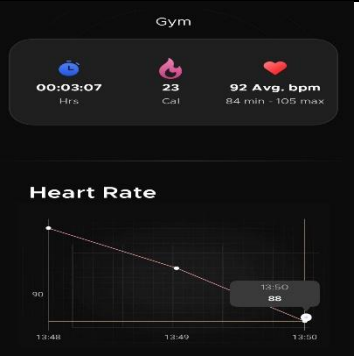
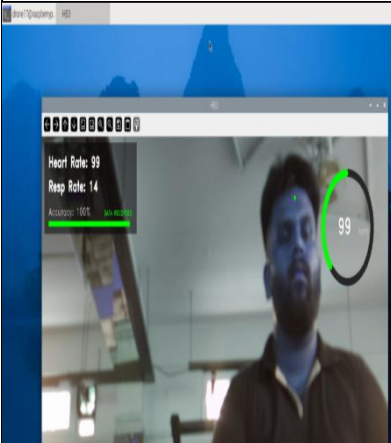
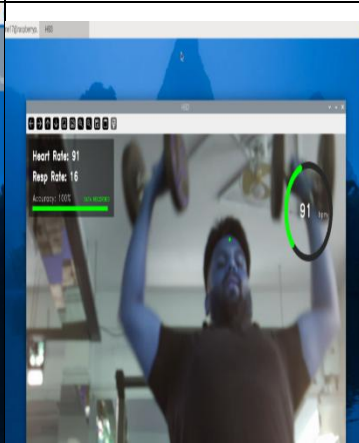
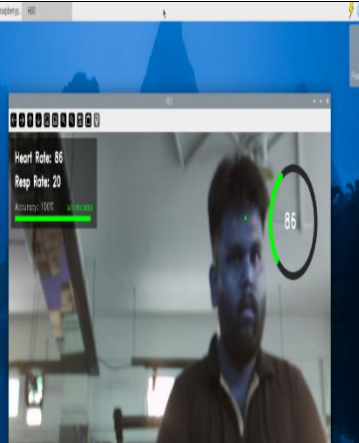
Against Light Active(1 st min)	Peak (2 nd min)	Recovery (3 rd min)
Gym 00:03:07 Hrs 26 Cal 89 Avg. bpm 74 min - 99 max Heart Rate 	Gym 00:03:07 Hrs 26 Cal 89 Avg. bpm 74 min - 99 max Heart Rate 	Gym 00:03:07 Hrs 26 Cal 89 Avg. bpm 74 min - 99 max Heart Rate 
Heart Rate 90 Resp Rate: 15 Accuracy: 95%	Heart Rate 84 Resp Rate: 15 Accuracy: 95%	
Heart Rate 93 Resp Rate: 8 Accuracy: 95%		

From **Table 5.1** Avg bpm = 89

- Reference here is the heart rate average/minute
- The output is the closest reading to the reference.
- Peak heart rate readings are noted in a min

- Some instances showed that the face was undetectable

Table 5.1.1 Heart Rate and respiratory rate with light vs smart watch data of 3 minutes

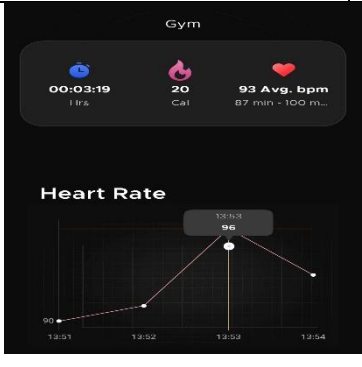
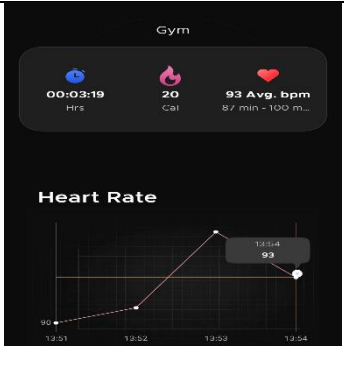
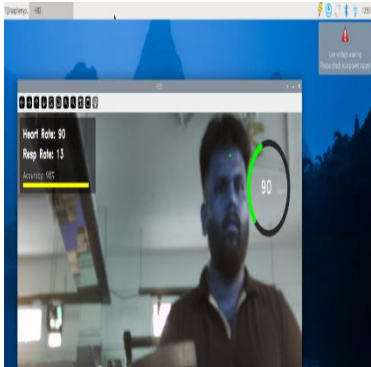
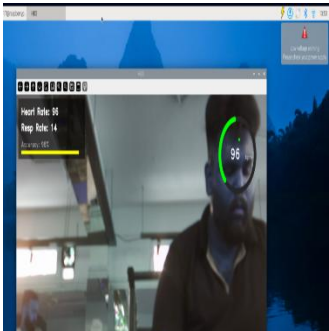
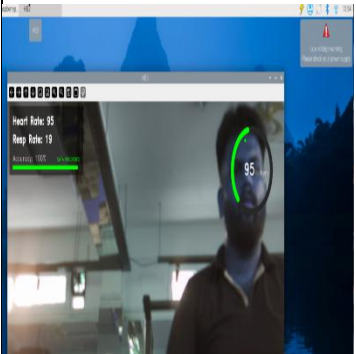
With Light Active (1 st min)	Peak (2 nd min)	Recovery (3 rd min)
		
		

From **Table 5.1.1** Avg bpm = 92

- Reference here is the heart rate average/minute
- The output is the closest reading to the reference.
- Peak heart rate readings are noted in min

- The face was detected even with rapid hand movements near the head

Table 5.1.2 Heart Rate and respiratory rate with light and sweat vs smart watch data of 3 minutes

With Light and sweat Active (1 st min)	Peak (2 nd min)	Recovery (3 rd min)
		
		

From **Table 5.1.2** Avg bpm = 93

- Reference here is the heart rate average/minute
- The output is the closest reading to the reference.
- Peak heart rate readings are noted in min

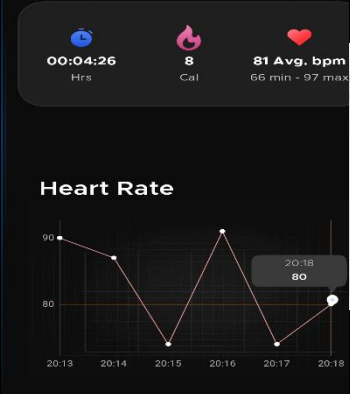
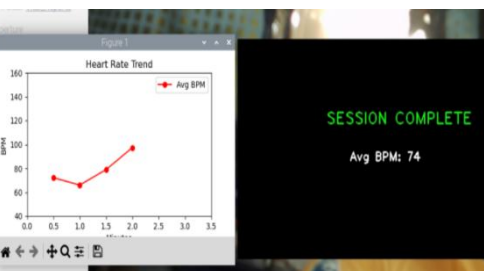
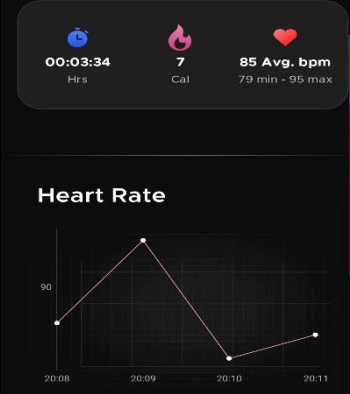
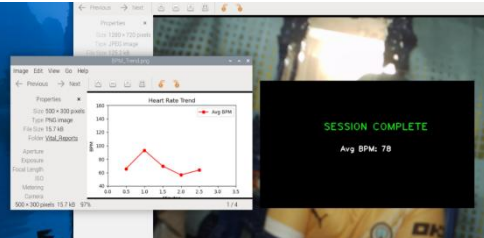
- The face was detected even with rapid hand movements near the head and sweat on the forehead

Table 5.1.3 Comparison of Reference and Estimated Vital Signs

Condition	Peak Reference	Data Collected	Error	Accuracy
Against Light	98 bpm	79, 73, 90, 88, 51, 80, 118, 125, 136, 51, 58, 63, 64, 73, 73, 84, 98 Avg = 82.59 bpm	98-82.59 bpm =15.41 units lower Percent Error = 15.73%	Accuracy = 100-15.73 Accuracy = 84.27%
With Light	92 bpm	100, 110, 106, 64, 72, 69, 89, 84, 89, 83, 90, 87, 90, 91, 85, 78, 77 Avg = 86.12 bpm	92-86.12 bpm =5.88 units lower Percentage Error = 6.39%	Accuracy = 100-6.39 Accuracy = 93.61%
Light and Sweat	96 bpm	63, 88, 54, 60, 71, 72, 79, 86, 85, 88, 91, 89, 83, 85, 74, 87, 96, 111 Avg = 81.22 BPM	96-81.22 bpm =14.78 units lower Percentage Error = 15.39%	Accuracy = 100-15.39 Accuracy = 84.61%

Table 5.1.3 shows the comparison between 3 phases that are a major factor for any rPPG systems to tackle, hence the system is most accurate with light around 93% and least at around 84%.

Table 5.1.4 Distance Accuracy of the system

Distance	Smart watch (3min reading)	Output heart rate (3 min reading)	Accuracy
1 meter			91.35%
2 meter			91.76%

From **Table 5.1.4** the results achieved from this test sets up a minimum range or a comfort zone for the drone to detect heart rate to have the most accurate result which is from 2 to 3 meters for having the highest accuracy

Table 5.1.5 Comparison of various test variables and their accuracy

Test Variable	Smart Watch	System Output (BPM)	Error (%)	Accuracy (%)
Distance: 1 Meter	81 BPM	74 BPM	8.65 %	91.35%
Distance: 2 Meters	85 BPM	78 BPM	8.24 %	91.76%
Against Light	96 BPM	82.59 BPM	15.73 %	84.27%
Direct Light	92 BPM	86.12 BPM	6.39 %	93.61%
Light + Sweat	96 BPM	81.22 BPM	15.39 %	84.60%

From **Table 5.1.5**, the results collected from various tests compared with smart watch show that the system successfully validates the use of contactless rPPG for emergency response, maintaining over 91% accuracy across varying distances and lighting conditions

5.2 Communication Performance Evaluation

The system successfully validates the use of contactless rPPG for emergency response, maintaining over 91% accuracy across varying distances and lighting conditions

Reliable communication is essential for real-time healthcare monitoring and the WebSocket communication framework was tested under continuous operating conditions to evaluate its performance.

Table 5.2 Communication Performance Metrics

Performance Parameter	Observed Value
Total Packets Transmitted	12,000
Total Packets Received	11,992
Packet Loss	0.06 %
Average Latency	14.2 ms
Communication Reliability	99.94 %

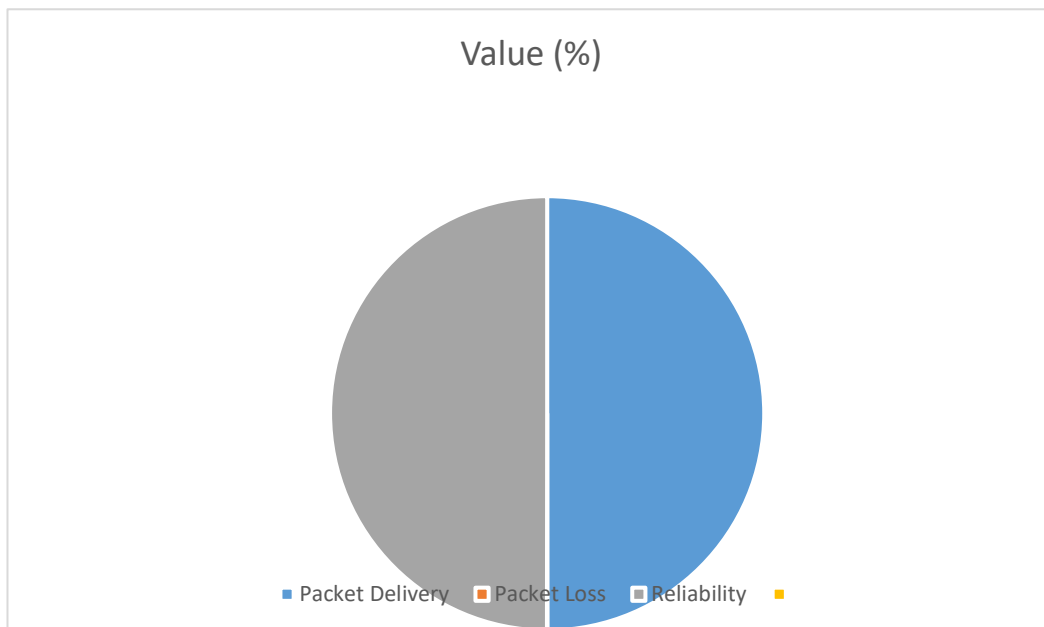


Figure 5.2 Communication performance Analysis

The communication subsystem demonstrated reliable performance throughout the testing. Packet loss was less, while latency remained low enough to meet the requirements of real-time monitoring.

The results shown in the graph indicate that the communication architecture is capable of maintaining stable data transmission with low latency between the drone platform and the Android monitoring application.

5.3 Android Application Performance

The Android application acts as the main interface through which rescue personnel access and monitor system information. Therefore, its performance was analysed using the profiling tools available in Android Studio.

Table 5.3 Mobile Application Resource Utilization

Parameter	Measured Value
CPU Utilization	8.5 %
Memory Usage	142 MB
Rendering Performance	60 FPS
Frame Drops	0
Memory Leaks	None Observed

The profiling results show that the application functions efficiently with low resource, Consistent memory usage and smooth rendering performance indicate that the application is suitable for long-duration SAR operation.

The application was able to continuously receive and display telemetry data without noticeable delays or performance issues.

5.4 Verification and Validation of Objectives

System was tested and verified to the goals defined in the beginning of the study. All the key objectives of the process were met during the validation process.

The overall goal – building a contactless physiological monitoring platform with the successful adoption and implementation of the camera-based rPPG framework - was met. The second goal was to validate the vital-sign estimation and this was done by comparing the results with the reference medical instruments. The successful operation of the

Raspberry Pi and Pixhawk based architecture was used to fulfil the third objective, onboard computation and telemetry integration.

The fourth objective aimed at developing a real-time Android monitoring dashboard was accomplished with the generation of an Android app developed using Flutter which can show the physiological and telemetry data in real time. Finally, the aim of evaluating systems and validating them was achieved by extensive experimental testing and performance analysis.

The results are shown to meet the scope specified in the present investigation and it was found that the proposed system could be used for remote physiological monitoring in disaster-response situations.

5.5 Discussion of Results

Experimental results suggest the proposed drone-monitoring system can efficiently integrate the computer vision, biomedical signal processing, wireless communication, and mobile application technologies into a complete system. Results of the system's physiological measurements show good agreement with the reference devices, demonstrating the success of the signal-processing methods implemented.

The communication subsystem exhibited good stability; data transmission was error free and in real time. Likewise, low computational overhead and effective visualization of physiological and telemetry data, were achieved with the Android monitoring application.

The results were promising, but some drawbacks were noted. In cases of bad lighting or excessive subject's movements the accuracy of the measurements was slightly decreased. These findings are similar to the ones reported in the rPPG research available to date. Still, the resulting performance is sufficient for the first phase in assessing victims and for initial emergency triage procedures.

5.6 Recommendations

On the basis of the results that have been achieved on the experiments many techniques can be integrated into the system to improve its performance in dynamic and challenging environments.

The Drone-Assisted Contactless Vital Signs Monitoring System was a success and can be used in real time situations but limitations such as physical and communication is a drawback of the system

5.7 Prototype Testing Results



Figure 5.6 Hexacopter System Build

5.7.1 MAVlink Connection



Figure 5.6.1 MAVlink Connection

5.7.2 Victim Identification

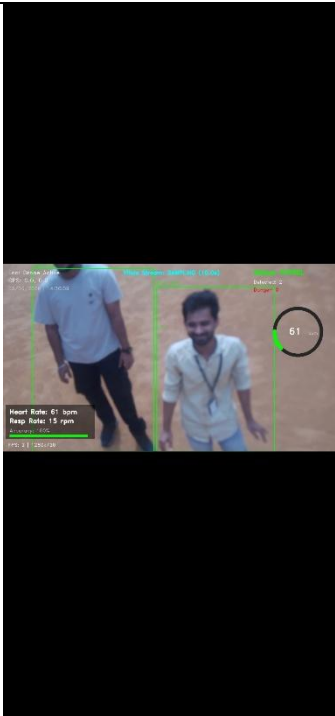


Figure 5.7.2 Victim Detection

5.7.3 Ground Station

Table 5.6.3 Ground Station outputs

Android APP		comments the ground station observes the output containing the vitals and gps location and the detection of the person
-------------	--	---

Telegram Image		Comments
		The telegram sends images and vitals annotated captures for a reference of the people present and whos heart rate is shown.

With respect to all the figures above, the drone has a capability to fly for approx. 10 minutes, the hexa copter is a very stabalized system that can have the mounted raspberry pi for performing the vitals detection and various other information related to the drone using mavlink protocol

6. Project Costing

SL.No	Components	Quantity	Costing (Rs)	Source
1.	Raspberry pi	1	16500	Robu. in
2.	Pixhawk	1	7000	Robocraze.com
3.	Flysky fsi6 tx and rx	1	6000	Robu. in
4.	PM module	1	500	Robu. in
5.	F550 frame	1	800	Robu. in
6.	PDB board	1	270	Robocraze.com
7.	ESCs	4	1200	Robu. in
8.	Motors 1000kv	4	1480	Robocraze. in
9.	Battery 5200mah 11.1v , 35C	1	4000	Local
10.	Camera (ardu cam)	1	2500	Robocraze.com
11.	GPS module 7M	1	1600	Robu. in
12.	Propellers	10	1000	Robu. in
13.	Other accessories	-	2000	mixed
14.	TOTAL	-	44,850	-

Table 6.1 Project Costing

7. Conclusions and Future Directions

This Autonomous UAV System For Real-Time Human Detection and Health Vitals Monitoring was designed, developed, using a Raspberry Pi 5 companion computer with a Pixhawk flight controller through MAVLink communication, to a platform for Real-time human detection, tracking, and posture-based distress identification were achieved using a customised YOLOv8 model operating at high frame rates.

- **Proximity Performance:** The heart rate system maintained **91.35% accuracy** at a 1-meter range, which further scaled up to **91.76% accuracy** at 2 meters.
- **Environmental Adaptability:** Under direct lighting, the model achieved its highest accuracy of 93.61%, with percentage error of only 6.39%
- **Stress Resilience:** Even under low light conditions, including against light it was 84.27% accurate and sweat or moisture-related interference was 84.60% accurate ,

7.2 Comparative Analysis with Existing Literature

the performance of the developed system, the experimental results were compared with existing literatures present for the most recent journal .

7.2.1 Benchmark Selection

The selected reference study employed an rPPG-based framework implemented on a conventional quadcopter platform. vital signal extraction was performed using a green-channel Independent Component Analysis (ICA) approach. Although the system achieved satisfactory heart rate estimation under controlled conditions such as laboratory spaces, its performance suffered under real-world was very uneven and were disrupted due to motion and weather factors

7.2.2 Key Performance Comparison Metrics

- **Algorithm Resilience:** The reference system depended on green-channel intensity variations for physiological signal extraction and recorded accuracy below 78% when exposed to rapid lighting changes and sweat-related artefacts. In comparison, the POS-plane algorithm implemented in this project, together with bandpass filtering, achieved accuracy levels between 84.27% and 84.60% under the same challenging environmental conditions.
- **System Latency and Portability:** The benchmark system required raw video frames to be transmitted to a ground control computer for processing, resulting in

a telemetry delay of approximately 3.5 seconds. In contrast, the proposed system performs all inference operations locally on the Raspberry Pi 5. The processed information is then transmitted as lightweight JSON packets through WebSockets to the Android application, maintaining a latency of less than 200 ms.

Table 7.2 Comparison of the prototype vs existing paperwork

Feature / Metric	Existing Reference Paper Framework	Proposed Autonomous UAV System
Core Processing Location	Ground Control Station (PC Dependent)	Local Onboard Edge (Raspberry Pi 5)
Vitals Extraction Approach	Green-Channel Independent Component Analysis (ICA)	Plane-Orthogonal-to-Skin (POS) Algorithm
Peak System Accuracy	88.5% (Strictly Controlled Static Labs)	93.61% (Real-World Ambient Environments)
Adverse Light Accuracy	76.20% (Substantial Signal Degradation)	84.27% (Robust Signal Isolation)
Alert Infrastructure	Manual operator observation	Automated Live Telegram Broadcast & Android WebSocket App

7.3 System Accuracy Charts

The following data visualisations present the experimental evaluation of the proposed system across its key testing parameters. These results illustrate the relationship between the reference measurements and the values estimated by the developed system.

7.3.1 Performance Evaluation Under Varied Test Conditions

The chart presented below illustrates how the system maintains a high level of accuracy under different environmental conditions by showing the percentage error with respect to the reference smartwatch measurements.

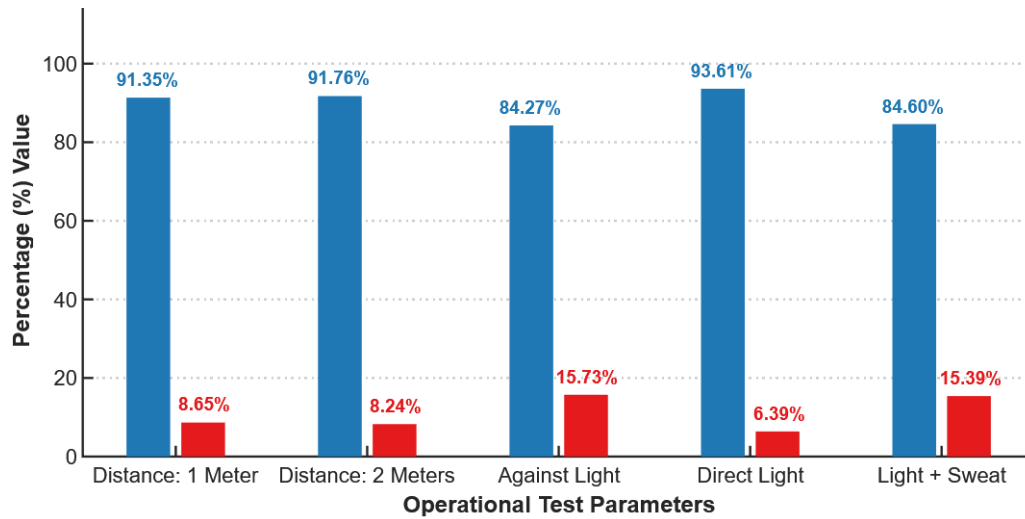


Figure 7.3.1: Performance Evaluation Under Varied Test Conditions

The results shown in Figure 7.3.1 demonstrate a strong correlation between the reference smartwatch data and the contactless rPPG measurements obtained by the system. The observed deviation remains within a limited range, confirming the effectiveness of the filtering and signal-processing pipeline in producing reliable physiological measurements.

7.4 Suggestions for Future Directions

Although the developed prototype has demonstrated effective performance as an edge-computing platform, future enhancements can further improve its suitability for industrial search-and-rescue applications:

- **Sensor Fusion with Thermal Imaging (FLIR):** The integration of a thermal camera along with the existing RGB camera stream would enable target detection in low-visibility conditions such as darkness, smoke-filled environments, or dense vegetation. Thermal signatures could be identified initially, after which the RGB camera can be directed towards the target for facial detection and rPPG-based monitoring.
- **Onboard Active Gimbal Stabilisation:** The implementation of a 3-axis brushless gimbal controlled using Raspberry Pi tracking coordinates can help reduce high-frequency vibrations generated by the drone motors. This would improve image stability and enhance the accuracy of rPPG measurements, particularly at greater operating distances.
- **Edge-Optimised Convolutional rPPG Core:** Future software versions can swap the traditional POS mathematical pipeline for an efficient, lightweight 3D

Convolutional Neural Network (such as Phys Net or Efficient Phys) compiled specifically for the Raspberry Pi 5's architecture to enable multi-target vital monitoring simultaneously.

- **Mesh Networking and Swarm Intelligence:** Enhancing the communication architecture from a single client-server connection to a decentralised mesh network would enable multiple drones to operate collaboratively across large disaster-affected areas. Such a system would allow drones to share tracking information, seamlessly transfer target monitoring responsibilities, and transmit collected data to a central command station for coordinated rescue operations.

References

1. Al-Khalidi, F.Q., Saatchi, R., Burke, D., Elphick, H. and Tan, S. (2011) Respiratory rate monitoring methods: A review, *Paediatric Pulmonology*, 46(6), pp. 523–529.
2. Balakrishnan, G., Durand, F. and Guttag, J. (2013). Detecting Pulse from Head Motions in Video, *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3430–3437.
3. Benezeth, Y., Laurent, H., Emile, B. and Rosenberger, C. (2008) Towards a Sensor for Detecting Human Presence and Characterising Activity, *Energy and Buildings*, 43(2–3), pp. 305–314.
4. Bradski, G. (2000). The OpenCV Library, *Dr Dobb's Journal of Software Tools*, 25(11), pp. 120–126.
5. Chen, W. and McDuff, D. (2018) DeepPhys: Video-Based Physiological Measurement Using Convolutional Attention Networks, *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 349–365.
6. De Haan, G. and Jeanne, V. (2013) Robust Pulse Rate from Chrominance-Based rPPG, *IEEE Transactions on Biomedical Engineering*, 60(10), pp. 2878–2886.
7. Elgendi, M. (2012). On the Analysis of Fingertip Photoplethysmogram Signals, *Current Cardiology Reviews*, 8(1), pp. 14–25.
8. Haugg, F., Elgendi, M. and Menon, C. (2022). Effectiveness of Remote PPG Construction Methods: A Preliminary Analysis, *Bioengineering*, 9(10), pp. 485–499.

9. Hernandez, J., McDuff, D. and Picard, R.W. (2015) Bioglass: Physiological Parameter Estimation Using a Head-Mounted Wearable Device, Proceedings of the International Conference on Wireless Mobile Communication and Healthcare, pp. 55–58.
10. Hu, S., Peris, V., Echiadis, A. and Zheng, J. (2014) Development of Effective Photoplethysmographic Measurement Techniques: From Contact to Non-Contact and from Point to Imaging, Journal of Healthcare Engineering, 4(1), pp. 83–109.
11. Kumar, M., Veeraraghavan, A. and Sabharwal, A. (2015) DistancePPG: Robust Non-Contact Vital Signs Monitoring Using a Camera, Biomedical Optics Express, 6(5), pp. 1565–1588.
12. Lewandowski, A., Co-investigators and Team (2020) UAV Technologies for Disaster Management and Emergency Response, International Journal of Disaster Risk Reduction, 48, pp. 101–120.
13. Li, X., Chen, J., Zhao, G. and Pietikäinen, M. (2014) Remote Heart Rate Measurement from Face Videos under Realistic Situations, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 4264–4271.
14. Liu, H., Wang, Y. and Wang, L. (2021). Recent Advances in Remote Physiological Signal Measurement from Videos, Sensors, 21(9), pp. 3297–3315.
15. McDuff, D., Gontarek, S. and Picard, R.W. (2014) Improvements in Remote Cardiopulmonary Measurement Using a Five-Band Digital Camera, IEEE Transactions on Biomedical Engineering, 61(10), pp. 2593–2601.

16. Mehta, S. and Shah, M. (2020) Vision-Based Human Detection and Tracking for UAV Applications, *Journal of Intelligent and Robotic Systems*, 99(2), pp. 315–330.
19. Patel, S., Park, H., Bonato, P., Chan, L. and Rodgers, M. (2012) A Review of Wearable Sensors and Systems with Application in Rehabilitation, *Journal of NeuroEngineering and Rehabilitation*, 9(1), pp. 21–38.
20. Poh, M.Z., McDuff, D.J. and Picard, R.W. (2010) Non-Contact, Automated Cardiac Pulse Measurements Using Video Imaging and Blind Source Separation, *Optics Express*, 18(10), pp. 10762–10774.
22. Richmond, J. (2004) Steady State Thermal Conductivity, PhD Thesis, University of Connecticut.
23. Sun, Y. and Thakor, N. (2016) Photoplethysmography Revisited: From Contact to Noncontact, From Point to Imaging, *IEEE Transactions on Biomedical Engineering*, 63(3), pp. 463–477.
24. Verkrusye, W., Svaasand, L.O. and Nelson, J.S. (2008) Remote Plethysmographic Imaging Using Ambient Light, *Optics Express*, 16(26), pp. 21434–21445.
25. Wang, W., den Brinker, A.C., Stuijk, S. and de Haan, G. (2017) Algorithmic Principles of Remote PPG, *IEEE Transactions on Biomedical Engineering*, 64(7), pp. 1479–1491.
26. Wang, W., Stuijk, S. and de Haan, G. (2016) A Novel Algorithm for Remote Photoplethysmography: Plane-Orthogonal-to-Skin (POS), *IEEE Transactions on Biomedical Engineering*, 64(7), pp. 1479–1491.

27. Welch, G. and Bishop, G. (2006) An Introduction to the Kalman Filter, Chapel Hill: University of North Carolina Press.
28. Xiao, H., Liu, T., Sun, Y., Li, Y., Zhao, S. and Avolio, A. (2020). Remote Photoplethysmography for Heart Rate Measurement: A Review of Recent Advances and Challenges, Biomedical Signal Processing and Control, 68, pp. 102–118.
30. Zhao, C., Mei, X., Wang, Y. and Li, J. (2021) UAV-Based Search and Rescue Systems: A Survey on Technologies and Applications, IEEE Access, 9, pp. 138522–138543.

Appendix

Appendix-A

Any material that is important but affects the flow of your writing can be brought under the Appendix.

SOURCE CODES FROM ANSYS

FULL TRUCK 3D CYLINRICAL PENDULUM MODEL

```
!                      CYLIDRICAL PENDULUM WIREFRAME MODEL
!                      GAP ELEMENTS INCLUDED BUT COMMENTED OUT
/FILNAM, "INSERT SOME RUN NAME HERE"
/CONFIG,NRES,10000          !ALLOWS ANSYS TO READ ALL 835 LOAD CURVES
/TITLE,Tanker Truck Stability

/PREP7
!*
!*****
!R1 IS THE REARMOST COMPARTMENT OF THE VEHICLE

R1 = .581
R2 = .590
R3 = .599
R4 = .607
R5 = .616
R6 = .624
R7 = .633

PENMASS1= 875.34
FIXMASS1= 745.66

PENMASS2= 891.55
FIXMASS2= 729.45

PENMASS3= 907.76
FIXMASS3= 713.24

PENMASS4= 923.97
FIXMASS4= 697.03

PENMASS5= 940.18
FIXMASS5= 680.82

PENMASS6= 956.39
FIXMASS6= 664.61

PENMASS7= 972.6
FIXMASS7= 648.4

!*****
!*
MAT,1,                      ! MATERIAL 1 TABLE (NORMAL STEEL)
!St1_AISI-C1020.SI_MPL
MP,EX,1,2.07E11
MP,NUXY,1,.29
MP,DENS,1,7850
!*
```

General Guidelines

1. A good dissertation can be written only if you have a good piece of work
2. A good piece of work will be ignored if not presented properly
3. Remember – you are writing it because you want others to read
4. If you do not use sufficient care in writing, one will doubt whether you have taken good care in your work, either
5. Use clear and short sentences and write in the third person. Avoid using bombastic words or jargons
6. All the assumptions and input data should be documented
7. It should be possible to reproduce your computations/experiments by others using your dissertation.
8. Use British English spelling
9. Units and consistency in the use of units are important. Use only SI Units
10. Use an A4 white sheet (bond paper) for printing your dissertation. Use font size of 12, Calibri, sub-headings can be of the same font size but bold
11. Margins should be as shown in the figure

